

# **AUTOMATIC SALINE LEVEL MONITORING SYSTEM USING IOT**

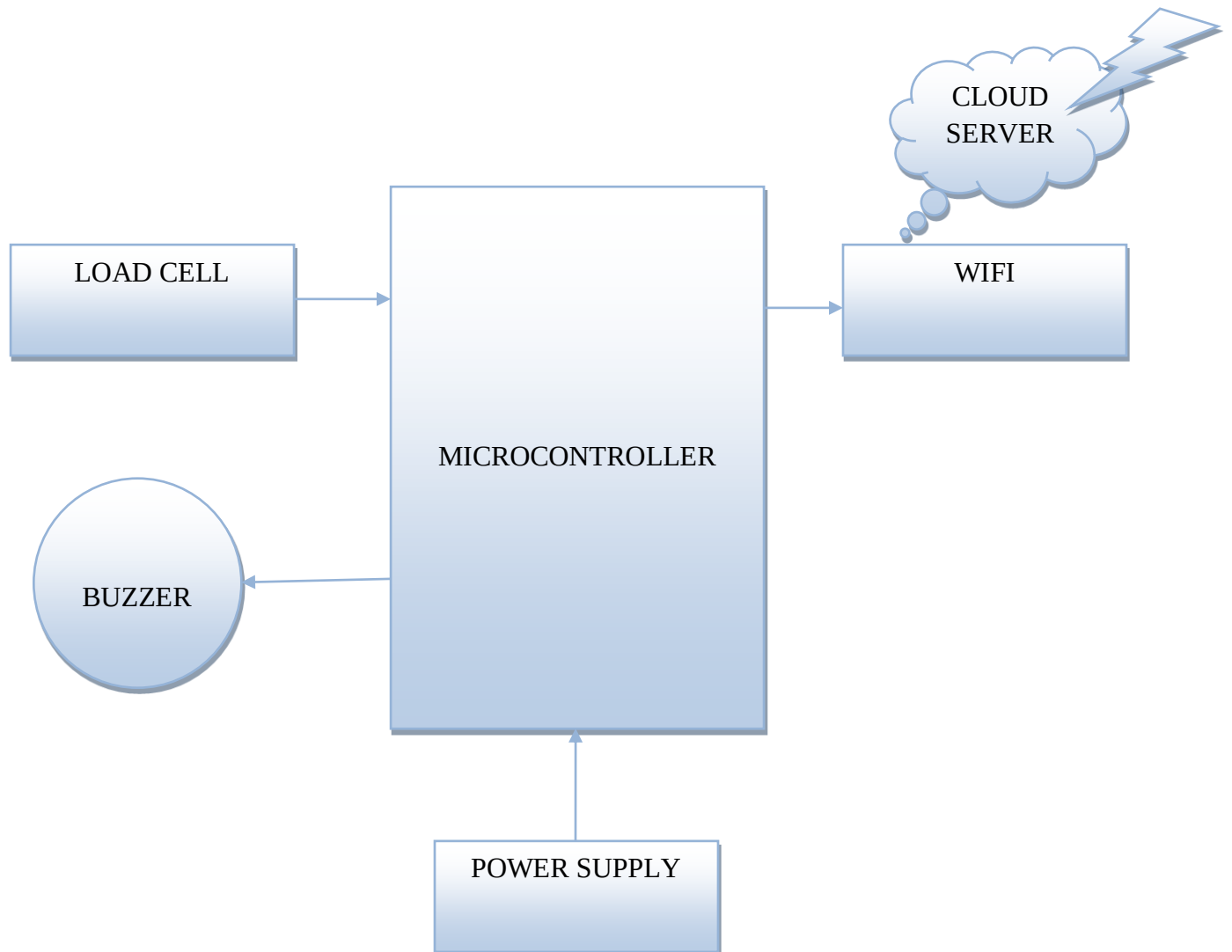
## **ABSTRACT**

An advanced medical technology called the Automatic Saline Level Monitoring System with IoT and Load Cell is intended to provide continuous, remote monitoring of the saline levels in IV bags. This system improves patient care and safety by giving medical professionals access to real-time saline level information through the combination of cutting-edge sensor technology and Internet of Things (IoT) connectivity. The saline bag's weight is precisely measured using a load cell, and a microcontroller transmits this information to an Internet of Things platform so that medical professionals can monitor saline levels from a distance using an intuitive interface. To ensure timely replenishment, alerts are generated when saline levels fall below a predetermined threshold. With considerations for calibration, data security, power management, and redundancy to ensure its dependability in a healthcare environment, the system provides a strong and secure solution that complies with healthcare regulations. This invention is expected to improve patient care and optimize medical operations in hospitals. Moreover, automated alert mechanisms are incorporated into the Automatic Saline Level Monitoring System. The system is designed to generate alerts when the saline level drops below a predefined threshold. This ensures that healthcare providers are notified promptly to replace the IV bag. By reducing the possibility of treatment disruptions, this feature enhances patient safety and care quality.

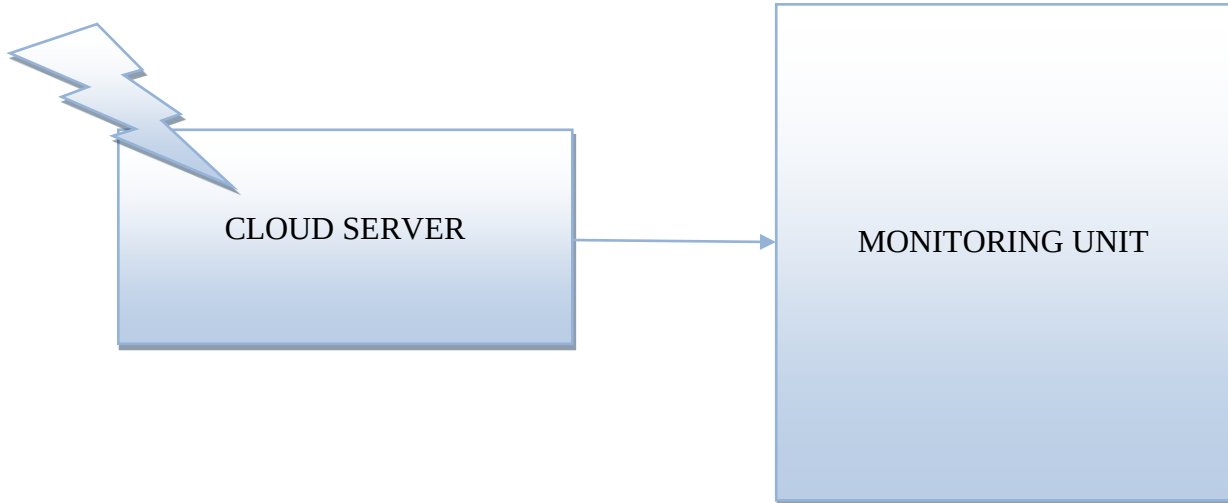
## INTRODUCTION

In order to improve patient care and streamline medical operations, the healthcare industry is always looking for new and innovative technologies. The Automatic Saline Level Monitoring System with IoT and Load Cell is one example of such innovation. This system ensures that patients receive the required treatment without interruption by addressing the critical need for continuous and remote monitoring of the saline levels in intravenous (IV) bags. For the sake of the patients' welfare, saline solutions must always be available. They are an essential part of medical treatment. In the past, healthcare professionals have calculated saline levels manually, which takes time, is prone to error, and necessitates frequent visits to the patient's bedside. The Automatic Saline Level Monitoring System uses a load cell, a high-precision sensor that can measure the weight of the saline bag, to try and change this practice. With this system's integration of IoT technology, real-time data transmission is possible, giving healthcare professionals prompt access to dependable and accurate saline level information. The microcontroller, which interfaces with the load cell and connects to the internet via Ethernet or Wi-Fi, is the brains behind the system. The system can send data to an IoT platform for visualization and monitoring thanks to this connectivity. Healthcare practitioners can make educated decisions and react quickly to any changes in saline levels by remotely accessing this data through intuitive dashboards or applications.

## HARDWARE SECTION:



**RECEIVER SIDE:**



## **SYSTEM REQUIREMENTS:**

### **HARDWARE REQUIRED:**

- MICRO-CONTROLLER
- BUZZER
- LOAD CELL
- CONNECTING WIRES
- SOLDERING KIT

### **SOFTWARE REQUIRED:**

- ARDUINO IDE
- EMBEDDED C

## **EXISTING SYSTEM**

The concept of saline flow monitoring and control in single bluetooth system has an disadvantage for monitoring multiple intravenous liquid flow access and control in a simultaneous process. The system is reliable, cost effective which has a saline monitoring transmitter parts with Arduino UNO as a major controller.

## **PROPOSED SYSTEM**

The deficiencies of traditional saline level monitoring techniques in healthcare facilities are intended to be addressed by the Automatic Saline Level Monitoring System using IoT and Load Cell. Saline levels in intravenous (IV) bags are continuously and remotely monitored thanks to a reliable and effective solution that makes use of cutting-edge sensor technology, Internet of Things connectivity, and intelligent data analysis.

The load cell, a precise sensor that can measure the weight of the saline bag with accuracy, is the central component of the system. The system tracks the saline level as it drops with usage by indirectly measuring the weight.

The control unit is a microcontroller, like an Arduino, Raspberry Pi, or specialized IoT development board. It processes data, communicates with the load cell, and oversees the system's operation.

The microcontroller can send data to an Internet of Things platform by connecting to the internet via Ethernet or Wi-Fi. Real-time and remote data accessibility is ensured by this connectivity.

## **LITERATURE SURVEY**

**TITLE:** Experimental Study on Blood Flow Rate Measurement using Intravascular Catheter

**AUTHOR:** Seng Sing Tan, Cheong Ming Lam, and Woan Eng Lim

### **DESCRIPTION:**

Measurement of blood flow rate is essential in assessing the cardiovascular status of patients with coronary artery diseases during treatments. It provides feedback and monitors any improvements in the vascular system after the cardiovascular surgery. This paper presents the latest development of a new approach to measure blood flow rate using a catheter involving the concept of heat exchanger. In the study, an intravascular catheter is inserted into a simulated blood stream and a heat transfer element attached at the distal end extracts a small amount of heat from blood to the cooling fluid in the heat transfer element. We can correlate the blood flow rate to the change of the outlet temperature of the cooling fluid. In this approach, we can determine the cardiac output without the need for introduction of saline solution or other substance into patient body. Unlike Swan-Ganz catheter, it does not alter the blood temperature. Consequently, it is repeatable without any possible adverse effects due to altering blood composition or heating in the vascular system. In this experimental study, we have done the comparison between different designs and settings, and confirmed that this approach has a great potential to be an effective way of measuring blood flow rate as, well as cardiac output.

**TITLE:** NonIntrusive Microwave System for Multiphase Flow Metering

**AUTHOR:** Aparna C. SheilaVadde, Vikram Melapudi, MN Suma Manoj Kumar  
KM

**DESCRIPTION:**

Accurate and reliable measurement of oil, gas and water flow rates as they flow along a pipe as a mixture continues to be a challenging problem in the oil and gas industry. Most of the solutions rely on radioactive sources. This paper explores the use of microwave sensors for flow rate measurement. Microstrip Patch sensors in transmission mode and a near field coaxial probe in reflection mode are used to estimate water fraction even in lossy (saline) medium using physics based models with minimal calibration. Reflection measurements from the Patch sensors are used to estimate gas fraction using empirical models which can enable measurement of gas in high loss cases. Gas velocity is measured from cross correlation. A low cost 5port measuring instrument was built to do reliable reflection and transmission measurements over a wide temperature range. The microwave system was tested in inhouse and external flow loops and in the field under varying temperatures and flow conditions and test results are presented in this paper.



**TITLE:** IN VIVO EVALUATION OF PRESSURE HEAD AND FLOW RATE ESTIMATION IN A CONTINUOUS-FLOW ARTIFICIAL HEART

**AUTHOR:** A. Tanaka<sup>1</sup>, M. Yoshizawa<sup>2</sup>, T. Yamada<sup>1</sup>, K. Abe<sup>1</sup>, H. Takeda<sup>3</sup>, T. Yambe<sup>4</sup>, S. Nitta<sup>4</sup>

**DESCRIPTION:**

To avoid using sensors with low biocompatibility and low durability in implantable TAH systems, the authors previously proposed a new method for estimating instantaneous values of flow rate and pressure head on the basis of voltage, current and rotational speed in a motor driven centrifugal pump. The previous in vitro experiments showed that the proposed estimator could automatically compensate for the effect of the change in blood viscosity on the estimation accuracy by employing two kinds of auto-regressive exogenous model. In this study, validity and reliability of this estimation method were ascertained in an acute animal experiment. In the experiment, two centrifugal blood pumps were implanted into an adult goat as a total artificial heart. Results of estimation were compared with true values when blood viscosity was changed by injecting physiological saline. The results indicated that the system could successfully estimate pressure head by compensating the change of viscosity although the estimation accuracy of the in vivo estimation was not so high as that of the previous in vitro tests.

**TITLE:** EFFECTS OF SALINEWATER FLOW RATE AND AIR SPEED ON LEAKAGE CURRENT IN RTV COATINGS

**AUTHOR:** Seog-Hyeon Kim, Reuben Hackam

**DESCRIPTION:**

Room temperature vulcanizing (RTV) silicone is increasingly being used to coat porcelain and glass insulators in order to improve their electrical performance in the presence of pollution and moisture. A study of the dependence of leakage current, pulse current count and total charge flowing across the surface of RTV on the flow rate of the saline water and on the compressed air pressure used to create the salt-fog is reported. The fog was directed at the insulating rods either from one or two sides. The RTV was fabricated from polydimethylsiloxane polymer, a filler of alumina trihydrate (ATH), a polymerization catalyst and fumed silica reinforcement, all dispersed in 1,1,1-trichloromethane solvent. The saline water flow rate was varied in the range 0.4 to 2.0 L/min. The compressed air pressure at the input of the fog nozzles was varied from 0.20 to 0.63 MPa. The air speed at the surface of the insulating rods was found to depend linearly on the air pressure measured at the inlet to the nozzles and varied in the range 3 to 14 m/hr. The leakage current increased with increasing flow rate and increasing air speed. This is attributed to the increased loss of hydrophobicity with a larger quantity of saline fog and a larger impact velocities of fog droplets interacting with the surface of the RTV coating.

**TITLE:** ESTIMATION OF CORONARY BLOOD FLOW BY CARDIAC THERMOGRAPHY IN OPEN CHEST CONDITIONS

**AUTHOR:** N. Gordon, S. Rispler, S. Sideman, R. Shofti, R. Beyar

**DESCRIPTION:**

We developed a thermographical method for quantitative estimation of coronary flow, by analyzing the dynamic response of the epicardial temperature to cold saline injection into the aortic root. In eight dogs, the heart was exposed and imaged by a thermographic camera. LAD flow was measured by a transit time flowmeter. End diastolic images were recorded for 20-30 sec after aortic root injection of 20cc of cold saline. Several LAD flow rates were achieved by one minute occlusion of the LAD, obtaining a reactive hyperemic response. A temperature response curve was obtained by averaging over an edge detected arterial segment for each frame. Several indices were calculated for each curve and correlated with coronary flow, one of which was a parameter of a heat transfer model, that was successfully fit to the curve. In conclusion, we developed a method for quantitative estimation of coronary blood flow by thermography, which can provide important information regarding coronary blood flow during open chest procedures.

## **HARDWARE DESCRIPTION:**

### **WeMos D1**

## **DESCRIPTION:**

The WeMos D1 WIFI ESP8266 Development Board is a small, affordable, and easy-to-use board that allows you to connect to the internet with your Arduino projects. WEMOS D1 WIFI is based on the ESP8266 chip, which is a powerful Wi-Fi microcontroller that can program using the Arduino IDE. The WeMos D1 WIFI ESP8266 Development Board is a great way to get started with IoT projects. WEMOS D1 WIFI is affordable, easy to use, and has a wide range of features. If you are looking for a way to connect your Arduino projects to the internet, the WeMos D1 board is a great option. This D1 WIFI ESP8266 Development Board is compatible with all the existing shields for Arduino because it has the same dimensions and pin layout as an Arduino UNO. This means that you can use any Arduino shield with the WeMos D1 board, without any modifications. This makes the WeMos D1 board a very versatile platform for IoT projects. You can use it to create a wide range of projects, using a variety of sensors and actuators. And because it is compatible with all the existing shields for Arduino, you can easily add new features to your projects. The WeMos D1 has a 32-bit processor that runs at 80 MHz, while the Arduino UNO has an 8-bit processor that runs at 16 MHz. This means that the WeMos D1 is significantly faster than the Arduino UNO. In terms of raw processing power, the WeMos D1 is about 10 times faster than the

Arduino UNO. This means that it can execute code much faster, which can be a big advantage for some applications.

## **FEATURES :**

The D1 WIFI ESP8266 has 160 KB of RAM, while the Arduino UNO has only 2 KB of RAM. This means that the WeMos D1 can store much more data in memory, which can be a big advantage for some applications. The WeMos D1 board has a number of features that make it well-suited for IoT projects, including: A built-in Wi-Fi module 11 digital I/O pins 1 analog input pin A Micro USB connector for power and programming A reset button A CH340G USB-to-UART chip WeMos D1 WiFi ESP8266 Specifications : – The D1 WIFI ESP8266 also has 4 MB of flash memory, while the Arduino UNO has only 32 KB of flash memory. This means that the WeMos D1 can store much more code in memory, which can be a big advantage for some applications. The D1 WIFI ESP8266 Development Board has the following specifications: Processor: ESP8266EX Clock speed: 80 MHz RAM: 160 KB Flash memory: 4 MB Operating voltage: 3.3 V Dimensions: 34.2 x 25.6 mm Pins: 11 digital I/O pins, 1 analog input pin Wi-Fi: 802.11 b/g/n Interfaces: Micro USB, I2C, SPI WeMos D1 WiFi ESP8266

**APPLICATIONS :**Overall, the WeMos D1 WIFI ESP8266 Development Board is a great option for anyone who wants to create IoT projects. It is affordable, easy to use, and compatible with a wide range of sensors and actuators. Here are some examples: Smart home devices Web servers Environmental monitoring Robotics IoT projects

## **REGULATOR**

### **DESCRIPTION**

A voltage regulator is designed to automatically maintain a constant voltage level. A voltage regulator may be a simple "feed-forward" design or may include negative feedback control loops. It may use an electromechanical mechanism, or electronic components.

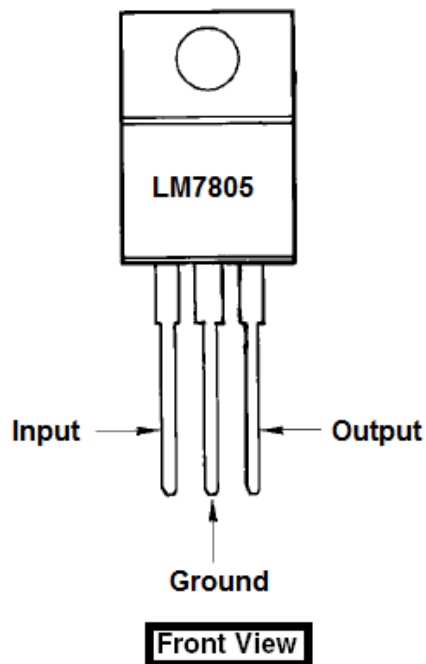
This series of fixed-voltage integrated-circuit voltage regulators is designed for a wide range of applications. These applications include on-card regulation for elimination of noise and distribution problems associated with single-point regulation. Each of these regulators can deliver up to 1.5 A of output current.



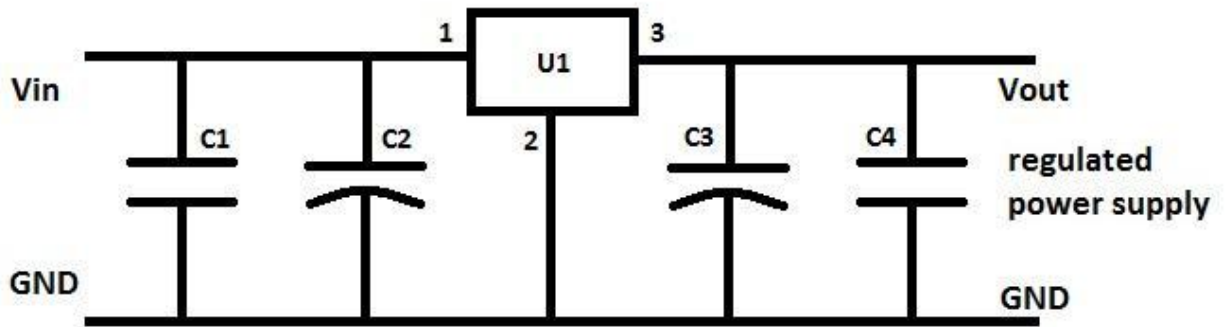
## **REGULATOR**

The internal current-limiting and thermal-shutdown features of these regulators essentially make them immune to overload. In addition to use as fixed voltage regulators, these devices can be used with external components to obtain adjustable output voltages and currents, and also can be used as the power-pass element in precision regulators.

The LM78XX series of three terminal positive regulators are available in the TO-220 package and with several fixed output voltages, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shut down and safe operating area protection, making it essentially indestructible.



## PIN DIAGRAM



## CIRCUIT DIAGRAM

If adequate heat sinking is provided, they can deliver over 1A output current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.

## FEATURES

- Output Current up to 1A
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24V
- Thermal Overload Protection
- Short Circuit Protection
- Output Transistor Safe Operating Area Protection

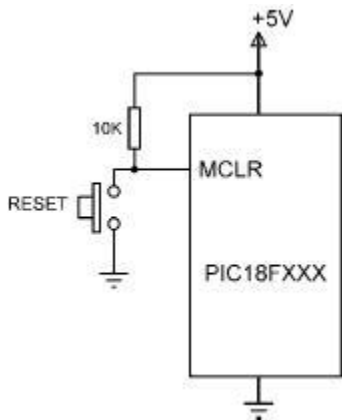


## RESET BUTTON

### DESCRIPTION

The first circuit pulls the RESET pin low to enable the chip. The second circuit is the same, but with a reset button. If you press that, the chip will get reset. When the 5V power is switched on, the capacitor shorts to 5V, and then gradually the RC circuit discharges to bring the reset pin to 0.

A power-on reset (PoR) generator is a microcontroller or microprocessor peripheral that generates a reset signal when power is applied to the device. It ensures that the device starts operating in a known state.



### CIRCUIT DIAGRAM FOR RESET BUTTON CONNECT WITH CONTROLLER

A power-on reset (PoR) is a circuit that provides a predictable, regulated voltage to a microprocessor or microcontroller with the initial application of

power. ... A PoR system can be a peripheral, but in sophisticated processors or controllers the PoR is integrated on the main chip.

## **BUZZER**

### **DESCRIPTION**

A buzzer or beeper is a signaling device, usually electronic, typically used in automobiles, household appliances such as a microwave oven, or game shows.

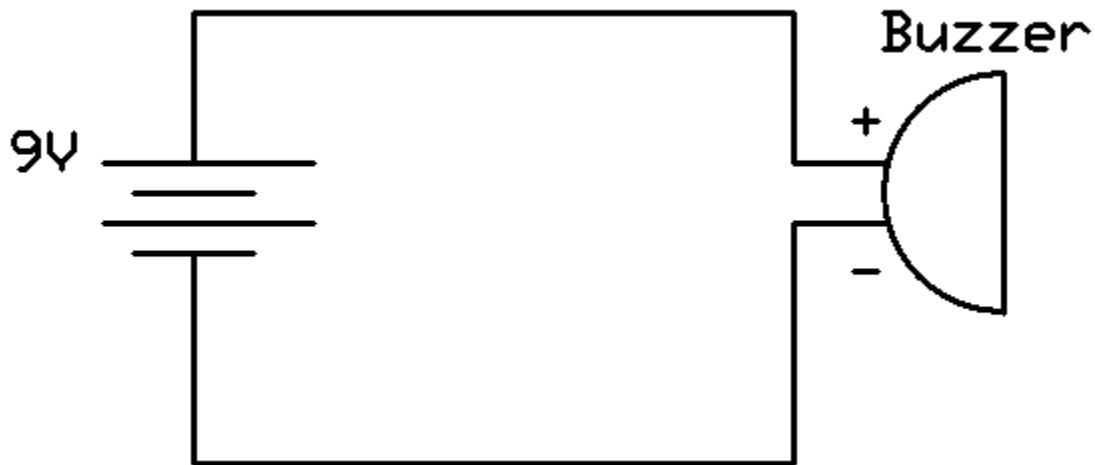


## **BUZZER**

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation of user input such as a mouse click or key stroke.

Buzzer is an integrated structure of electronic transducers, DC power supply, widely used in computers, printers, copiers, alarms, electronic toys, automotive

electronic equipment, telephones, timers and other electronic products for sound devices. Active buzzer 5V Rated power can be directly connected to a continuous sound, this section dedicated sensor expansion module and the board in combination, can complete a simple circuit design, to "plug and play."



### **CIRCUIT DIAGRAM**

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke.

It generates consistent single tone sound just by applying D.C voltage. Using a suitably designed resonant system, this type can be used where large sound volumes are needed. At Future Electronics we stock many of the most common types categorized by Type, Sound Level, Frequency, Rated Voltage, Dimension and Packaging Type.

## **FEATURES**

- Input supply: 5 VDC
- Current consumption: 9.0 mA max.
- Oscillating frequency:  $3.0 \pm 0.5$  KHz
- Sound Pressure Level: 85dB min

## **APPLICATIONS**

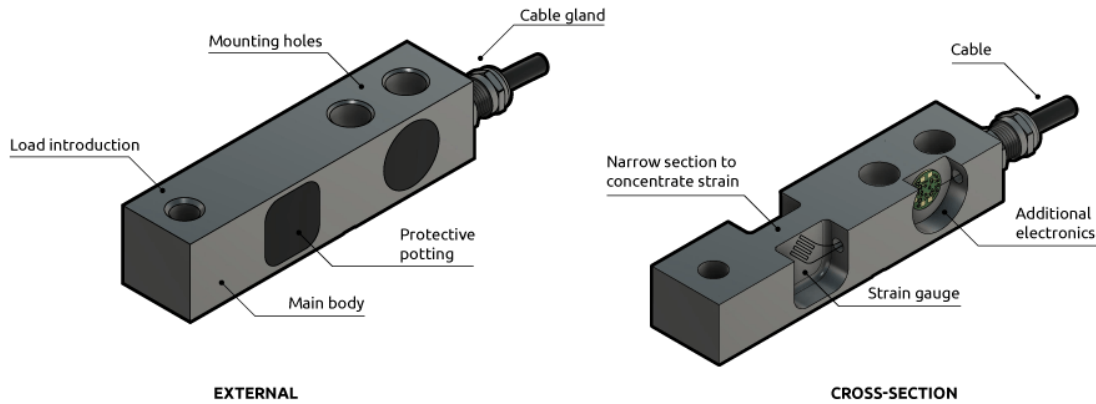
- Confirmation of user input (ex: mouse click or keystroke)
- Electronic metronomes
- Sporting events
- Judging Panels
- Annunciator panels

# **LOAD CELL**

## **DESCRIPTION**

A load cell is an electro-mechanical sensor used to measure force or weight. It has a simple yet effective design which relies upon the well-known transference between an applied force, material deformation and the flow of electricity. They are incredibly versatile devices that offer accurate and robust performance across a diverse range of applications. It's no surprise that they have become essential to many industrial and commercial processes, from automating car manufacturing to weighing your shopping at the checkout. As technology explodes forward, many new and exciting applications are emerging that also stand to benefit from using load cells. New advances in robotics, haptics and medical prostheses, to name a few, all need effective ways to measure forces and weights. New types of load cells are continuously being designed to meet the needs of this ever-changing market.

## Load-cell anatomy



When we use load cells, one end is usually secured to a frame or base, while the other end is free to attach the weight or weight-bearing element. When force is applied to the body of the load cell, it flexes slightly under the strain. This is similar to what happens to a fishing rod when a fisherman hooks a fish. The fisherman will secure the rod in their hands while the fish applies a pulling force on the other end of the fishing line. The result of this force is that the fishing rod bends, with a bigger, stronger fish, causing the bend to be more extreme. When this action happens to a load sensor, the deformation is very subtle and not visible to the naked eye. To measure the deformation, strain gages are tightly bonded to the body of the load cell at pre-determined points, causing them to deform in unison with the body. The resulting movement alters the electrical resistance of the strain gages in proportion to the amount of deformation caused by the applied load. Using signal conditioning electronics, the electrical resistance of the strain gages

can be measured with the resulting signal being output as a weight or force reading.

## **TECHNOLOGY OF A LOAD CELL**

A typical load cell consists of two parts: the main body and an attached electrical circuit. The main body is what bears the weight or force and accounts for most of the load cell's size. Typically, it is made from high-grade steel or aluminium, which ensures mechanical reliability, and predictable and uniform strain distribution. The electrical circuit is housed within the load cell, tightly bonded to the main body. The circuit includes strain-gauges which are specialised parts of the circuit designed to sense the deformations of the main body. These strain-gauges consist of thin, electrically conductive wire or foil arranged in a tight zig-zag pattern. This pattern makes them sensitive to stretch and compression along their length, but insensitive across their width. As such, they can be precisely positioned to sense forces that run along particular axes. For example, shear beam load cells have their strain gauges positioned at a 45-degree angle to the loading axis, so as to maximise the detection of the shear strain running through the load cell.

## **ADAPTER (12V 1AMP)**

### **DESCRIPTION**

An AC adapter, AC/DC adapter, or AC/DC converter is a type of external power supply, often enclosed in a case similar to an AC plug. Adapters for battery-powered equipment may be described as chargers or rechargers (see also battery charger). AC adapters are used with electrical devices that require power but do not contain internal components to derive the required voltage and power from main power. The internal circuitry of an external power supply is very similar to the design that would be used for a built-in or internal supply.

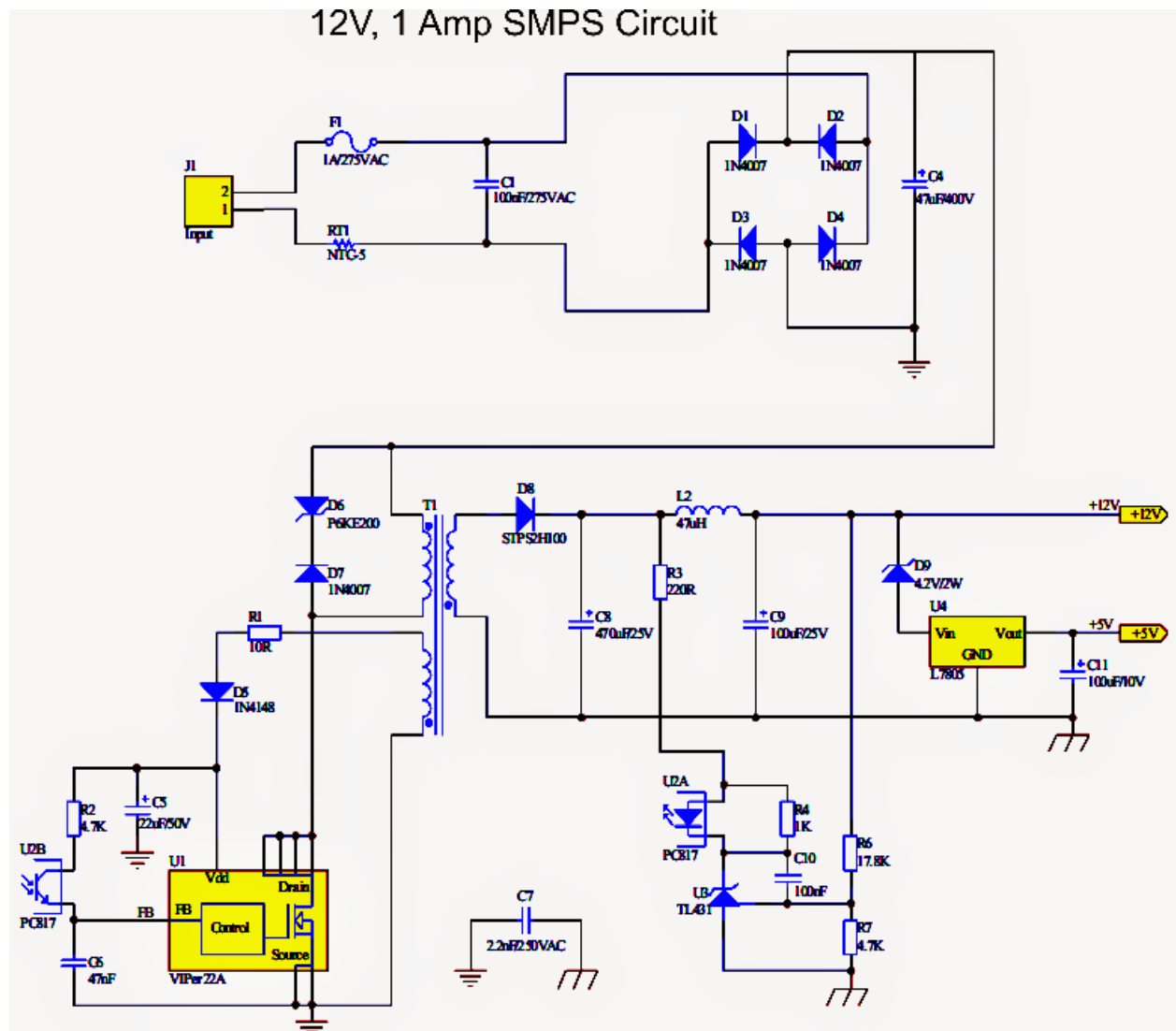


## **ADAPTER (12V 1AMP)**

An adapter is a device that converts attributes of one electrical device or system to those of an otherwise incompatible device or system. Some modify power or signal attributes, while others merely adapt the physical form of one electrical connector to another. In a computer, an adapter is often built into a card that can be inserted into a slot on the computer's motherboard. The card adapts



information that is exchanged between the computer's microprocessor and the devices that the card supports.



## CIRCUIT DIAGRAM

An electric power adapter may enable connection of a power plug, sometimes called, used in one region to a AC power socket used in another, by offering connections for the disparate contact arrangements, while not changing the voltage. An AC adapter, also called a "recharger", is a small power supply that

changes household electric current from distribution voltage) to low voltage DC suitable for consumer electronics.

Some modify power or signal attributes, while others merely adapt the physical form of one electrical connector to another. For computers and related items, one kind of serial port adapter enables connections between 25-contact and nine-contact connectors, but does not affect electrical power- and signalling-related attributes

## **FEATURES**

- Output current:1A
- Supply voltage: 220-230VAC
- Output voltage: 12VDC
- Reduced costs
- Increased value across front-office and back-office functions
- Access to current, accurate, and consistent data
- It generates adapter metadata as WSDL files with J2CA extension.

## **APPLICATIONS**

- Back-end systems which need to send purchase order data to oracle applications send it to the integration service via a integration server client.
- SMPS applications.

## SOFTWARE DESCRIPTION:

### ARDUINO IDE:

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

Programs written using Arduino Software (IDE) are called **sketches**. These sketches are written in the text editor and are saved with the file extension `.ino`. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

Before uploading your sketch, you need to select the correct items from the **Tools** > **Board** and **Tools** > **Port** menus. The boards are described below. On the Mac, the serial port is probably something like `/dev/tty.usbmodem241` (for an Uno or Mega2560 or Leonardo) or `/dev/tty.usbserial-1B1` (for a Duemilanove or earlier USB board), or `/dev/tty.USA19QW1b1P1.1` (for a serial board connected with a Keyspan USB-to-Serial adapter). On Windows, it's probably COM1 or COM2 (for

a serial board) or COM4, COM5, COM7, or higher (for a USB board) - to find out, you look for USB serial device in the ports section of the Windows Device Manager. On Linux, it should be `/dev/ttyACMx` , `/dev/ttyUSBx` or similar. Once you've selected the correct serial port and board, press the upload button in the toolbar or select the **Upload** item from the **Sketch** menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Diecimila) that lack auto-reset, you'll need to press the reset button on the board just before starting the upload. On most boards, you'll see the RX and TX LEDs blink as the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error.

When you upload a sketch, you're using the Arduino **bootloader**, a small program that has been loaded on to the microcontroller on your board. It allows you to upload code without using any additional hardware. The bootloader is active for a few seconds when the board resets; then it starts whichever sketch was most recently uploaded to the microcontroller. The bootloader will blink the on-board (pin 13) LED when it starts (i.e. when the board resets).

## **PROGRAMMING LANGUAGE:**

Embedded C is a set of language extensions for the C programming language by the C Standards Committee to address commonality issues that exist between C extensions for different embedded systems.

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programming requires nonstandard extensions to the C language in order to support exotic features such as fixed-point arithmetic, multiple distinct memory banks, and basic I/O operations

An embedded system is a computer system with a dedicated function within a larger mechanical or electrical system, often with real-time computing constraints. It is embedded as part of a complete device often including hardware and mechanical parts. Embedded systems control many devices in common use today. Ninety-eight percent of all microprocessors are manufactured as components of embedded systems.

Examples of properties of typical embedded computers when compared with general-purpose counterparts are low power consumption, small size, rugged operating ranges, and low per-unit cost. This comes at the price of limited processing resources, which make them significantly more difficult to program and to interact with. However, by building intelligence mechanisms on top of the hardware, taking advantage of possible existing sensors and the existence of a network of embedded units, one can both optimally manage available resources at the unit and network levels as well as provide augmented functions, well beyond those available. For example, intelligent techniques can be designed to manage power consumption of embedded systems.

Modern embedded systems are often based on microcontrollers (i.e. CPU's with integrated memory or peripheral interfaces), but ordinary microprocessors (using external chips for memory and peripheral interface circuits) are also common, especially in more-complex systems. In either case, the processor(s) used may be types ranging from general purpose to those specialized in certain class of computations, or even custom designed for the application at hand. A common standard class of dedicated processors is the digital signal processor (DSP).

Since the embedded system is dedicated to specific tasks, design engineers can optimize it to reduce the size and cost of the product and increase the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, and largely complex systems like hybrid vehicles, MRI, and avionics. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside a large chassis or enclosure.

## EMBEDDED C

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## **SYSTEM TESTING**

### **HARDWARE TESTING**

An embedded system is a computer system designed to do one or a few dedicated and/or specific functions often with real-time computing constraints." So embedded system is nothing but low cost, low space and low maintenance system. It is designed to handle a specific task to be controlled by one or more main processing cores that are typically either microcontrollers or digital signal processors.

### **TESTING**

Testing is very vague term and to define it we should know and we should have some baselines. Basically to understand testing we should know few things:

1. Testing is not independent module it is interlinked with the system to be tested.
2. To test the system you should know the system, this actually means if you want to test anything first you should have a thorough knowledge of the system.

These are very basic things and can be applied to any system/scenario. For Embedded Systems testing we should be aware of the testing environment.

We have basically three categories for testing:

1. White box testing
2. Black Box testing
3. Grey Box testing

## **WHITE BOX TESTING**

It is done with the actual code. If we are using some API's any other tool to see code coverage etc you are doing white box testing. If we are modifying the code to see whether it is working as expected is not, is white box testing.

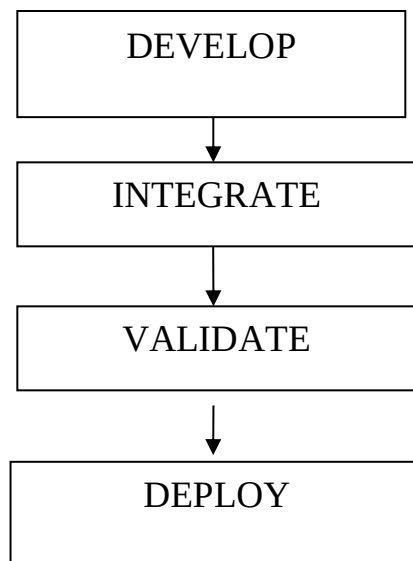
## **BLACK BOX TESTING**

If we believe that every system has a bug and tries to find defects without even knowing the internal coding of the system.

## **GREY BOX TESTING**

If we know how internal coding has been done and writing the test cases based on how the system will be used comes under grey box testing.

## **SOFTWARE TESTING**



## **development step**

### **EMBEDDED SOFTWARE IS WRITTEN FOR SPECIFIC HARDWARE**

- Requires specific inputs and outputs
- Target has different architecture
  - Endianness
  - Memory model
  - Hardware accelerators
- Target has limited resources
  - Memory
  - Disk

### **SIMULATION OVERCOMES LIMITED ACCESS TO HARDWARE**

- Different levels of simulation
  - Emulation: qemu (PowerPC, ARM)
  - Virtualization: KVM, Virtual Box
  - Stubbing/hardware abstraction
- PC has much more resources and performance

### **TESTING AND OPEN SOURCE SOFTWARE**

- Open source tools to support testing
  - Unit tests
  - Doctest
  - D-Bus

- Testing of open source tools
  - Gstreamer
  - Linux kernel

## **LINUX KERNEL HAS STUBS AND TEST FRAMEWORK**

- For most device types, stub implementation exists
  - For USB gadget: dummy\_hcd, zero
  - For MTD (= flash): block2mtd
- Linux Test Project (LTP) offers a large test suite
  - Mainly for file systems and networking.
  - Mainly regression, load and performance tests.

Developer should write tests from the start, It saves time, Share the tests with other developers, Even more important for open source projects, For embedded systems, hardware abstraction / stubs are essential.

## **TEST DESIGN TOOLS**

Tools that help you decide what tests need to be executed. Test data and test case generators.

## **GUI TEST DRIVERS**

Tools that automate execution of tests for products with graphical user interfaces. Client/server test automation tools, including load testers, also go here.

## **LOAD AND PERFORMANCE TOOLS**

Tools that specialize in putting a heavy load on systems (especially client-server Systems). These tools are often also GUI test drivers.

## **TEST MANAGEMENT TOOLS**

Tools that automate execution of tests for products without graphical user interfaces. Also tools that help you work with large test suites.

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